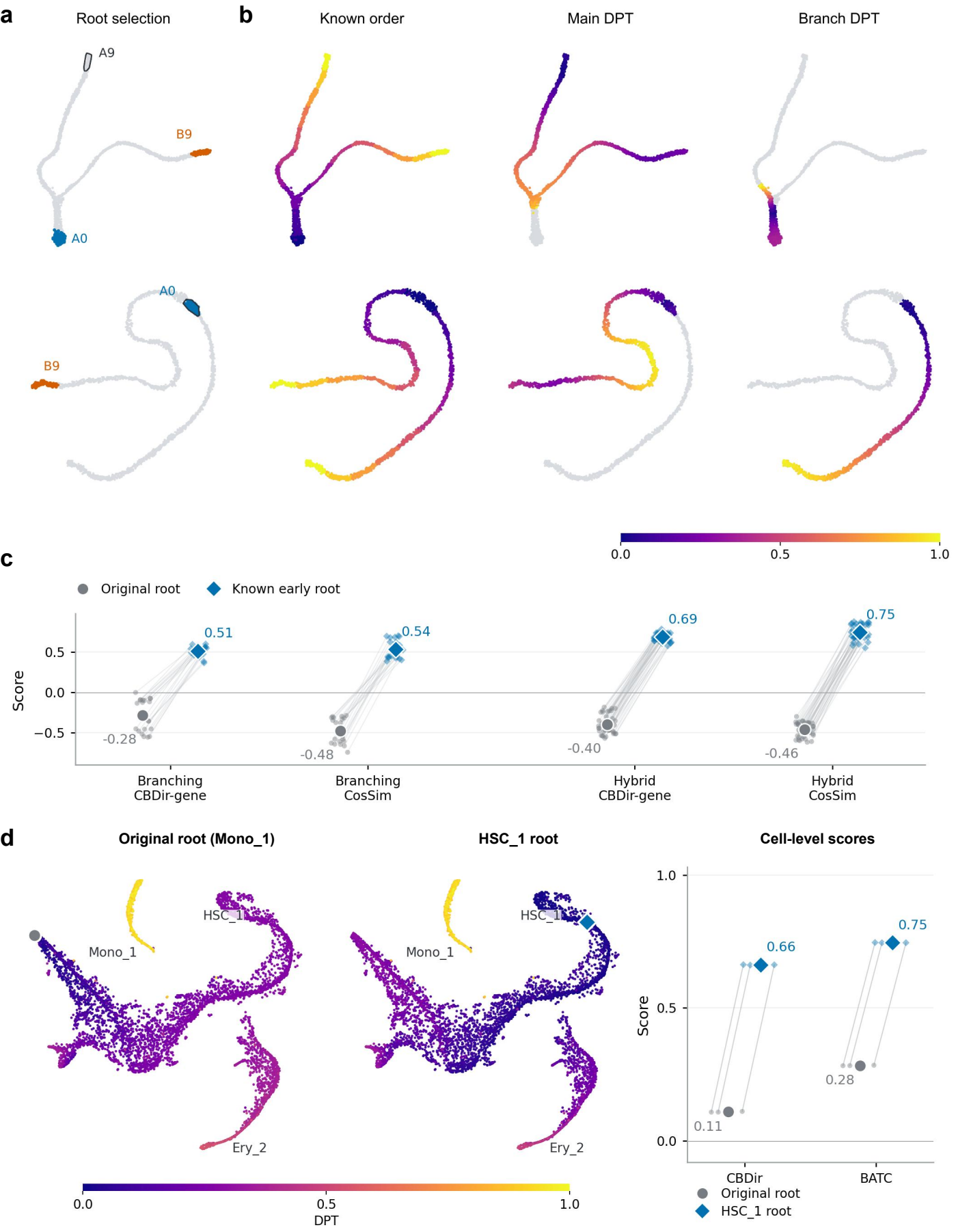

a
Root selection
A9
B9
A0
A0
B9
b
Known order
Main DPT
Branch DPT
0.0
0.5
1.0
c
Original root
Known early root
Score
0.5
0.0
−0.5
0.51
0.54
0.69
0.75
-0.28
-0.48
-0.40
-0.46
Branching CBDir-gene
Branching CosSim
Hybrid CBDir-gene
Hybrid CosSim
d
Original root (Mono_1)
HSC_1 root
Cell-level scores
HSC_1
Mono_1
Ery_2
HSC_1
Mono_1
Ery_2
0.0
0.5
1.0
DPT
Score
1.0
0.5
0.0
0.66
0.75
0.11
0.28
CBDir
BATC
Original root
HSC_1 root